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Technology-enhanced language learning in English language education: Performance analysis, core publications, and emerging trends

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ABSTRACT

As technology use has become the norm in education, this bibliometric analysis of technology-enhanced language learning (TELL) aims to reveal its current state-of-the-art and emerging trends. Analysis of 1,816 publications (1,745 articles and 71 reviews) from Web of Science demonstrated growing interests in the field and core publications in the field. Bibliographic coupling identified eight research fronts, with a particular emphasis on the established flipped learning (FL) pedagogy and expanding influence of mobile assisted language learning (MALL) and digital game-based learning (DGBL). These approaches are at the forefront in shaping English language skill acquisition, especially in writing, with the rise of technology multimodality and informal digital learning as nascent yet significant areas for future research. Anchored in the bioecological model, the research highlights the integral role of student outcomes across various competencies influenced by systemic factors. The study stresses the necessity for education stakeholders to blend technology with pedagogical strategies, a need further accentuated by the COVID-19 pandemic. The study's major contribution lies in its comprehensive synthesis of TELL's current landscape and for both future research and education endeavours in the field of English TELL.

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Introduction

Technology has become increasingly prevalent in language education (Palacios Hidalgo, 2020), as evident with dedicated subfields from computer-assisted language learning (CALL; Gillespie, 2020), mobile-assisted language learning (MALL; Elaiish et al., 2019), to technology-enhanced language learning (TELL; Shadiev & Yang, 2020). In addition to the wealth of technologies investigated (Lim & Aryadoust, 2021; Zhang & Zou, 2022b), studies have demonstrated positive student perceptions and impact on language learning (Goksu et al., 2022), enhancing motivation, engagement, and confidence (Shadiev & Wang, 2022; Wei, 2022), and positive outcome in both receptive skills in vocabulary, grammar, listening, and reading (Zhang & Zou, 2022a) and productive skills of speaking and writing (Shadiev & Yang, 2020). In particular, as the main international language for communication, learning English as a foreign language (EFL) has dominated research in technology use in language learning (Goksu et al. 2022; Shadiev & Yang, 2020).

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Despite decades of calls for stronger integration of technology in language education (Dede, 2000; Hubbard, 2013), the concept of *emergency remote teaching* during the recent COVID pandemic accentuated the difficulties and challenges faced by educators (Atmojo & Nugroho, 2020; Moorhouse et al., 2021). As the pandemic has resided and the role of technology becomes more important than ever, there is a need for educators and researchers to be well-prepared with an up-to-date overview of the field. This study therefore aims to identify the latest trends in technology-enhanced English education, providing a reference to inform nascent and veteran scholars on the latest developments and topics for future research. Findings from the bibliometric analysis are further framed within the bioecological model (Bronfenbrenner, 1979; Bronfenbrenner & Morris, 2006) to illustrate the holistic landscape of the field.

Technology in English education

Rapid technological advances have blurred the lines between the previous notions of specific technology use in language education (Dooly & Masats, 2015; Palacios Hidalgo, 2020; X. Chen et al., 2021; Wei, 2022; Zhang & Zou, 2022b). They work in tandem to deliver a comprehensive educational experience. In fact, beyond earlier specialised notions of MALL and CALL, TELL encompasses the entire spectrum, from online learning, distance education, distributed education, virtual environment, learning management systems, the internet and web 2.0, and massive open online courses (MOOC), to synchronous and asynchronous teaching and learning in e-learning, and flipped, blended, or hybrid learning. Use of any digital technology in formal or informal learning both inside and outside the classroom can thus fall under the umbrella term of TELL (Marijuan & Sanz, 2017).

Application of technology has demonstrated a positive impact on language learning, whether focusing on overall or specific technologies. For instance, Zhang and Zou (2022a) looked at 41 publications from 2009 to 2020 and identified computers, mobile devices, printed materials, audio players, and PowerPoint slides as the five main tools of multimedia technologies in supporting vocabulary, listening, reading, and grammar. In another review, the authors (2022b) investigated 57 publications in 10 journals until 2019 and identified mobile devices, multimedia, speech-to-text and text-to-speech, and digital game-based learning as the top five technologies used in language teaching. On the other hand, Shadiev and Yang (2020) found that the most used were digital games and online videos, amongst the 23 technologies in the top ten technology and language learning journals of 398 publications between 2014 and 2019. In terms of skills, (Elaish et al., 2019) focused on mobile-assisted English learning and found that vocabulary was the most-used skill, and that motivation was the most identified problem in their review of 69 publications from 2010 to 2015. The literature underscored not only the diverse profusion of technology, but also their positive impact on language learning. In short, the ubiquitous and authentic nature of technology promotes active, flexible, efficient, individualised, and motivating learning processes that cater to individual learners' ability, preference, and learning styles (Palacios Hidalgo, 2020).

Bioecological approach to technology use in English language education

Drawing on Bronfenbrenner's (1979) bioecological systems theory, a person's development is impacted by the dynamics between the contexts of five environmental systems (Bronfenbrenner & Morris, 2006). At the centre is the microsystem, where interactions between the person and the immediate people (i.e., teacher or parents) or objects (i.e., books or toys) take place. The connections between different parts of the microsystem constitute the mesosystem (i.e. teacher and parent interaction). The exosystem and macrosystem comprise broader indirect environmental impacts such as schools or governments for the former, and nations or cultural values for the latter. Finally, the chronosystem refers to the impact of timely events on the entire ecology, such as the recent COVID pandemic.

Modification to the bioecological system has been proposed with the increasing use of technology. Johnson and Pupilampu (2008) introduced the techno-subsystem within the microsystem, where learning outcome is impacted by the types of technology and usage. Previous findings have shown the

importance of technology within the system, wherein compared to the microsystem of socioeconomic status, the techno-subsystem of internet usage at home was found to exert more influence on cognitive development (Johnson, 2010). Chiu (2020) further validated the impact of technology use in the ecological model using structural equation modelling and highlighted the importance of outside-school technology use in increasing the effect of inside-school use on learning outcome. More recently, Navarro and Tudge (2023) introduced neo-ecological theory as a conceptual framework to understand the impact of technology on learners in the digital age. The authors proposed that the individual can co-exist within both the virtual (online) and physical (face-to-face) setting in the microsystem. Herein, the traditional nature of the microsystem is no longer spatial but relational; the virtual microsystem provides a phenomenological experience characterised by the individuals' engaging and disengaging with the virtual platform. Access to digital technology is further influenced by macrosystemic factors that highlight class (Navarro & Tudge, 2023) and economic inequalities (Chiu, 2020) that adversely affect the virtual and physical environments. In short, the use of technology needs to be configured around the students while considering other variables.

Research questions

The rapid emergence and widespread adoption of new technologies have led to a significant increase in related publications, necessitating frequent updates (Goksu et al., 2022; X. Chen et al., 2021) to provide knowledge consolidation and inform scholars on future research directives (Marijuan & Sanz, 2017). Nevertheless, with the overlapping nature of technology use, focusing on specific aspects such as CALL or MALL may fail to present a thorough review of the field (Yilmaz et al., 2022). This is further aggravated when applying bibliometric approaches to small publication sample sizes ($n < 350$) and/or specific journals (X. Chen et al., 2021).

To address these limitations and provide a more holistic understanding, this study complements the existing literature by taking different bibliometric approaches. First, the search terms have been generated by drawing on recent developments and previous reviews. Second, local citation count was employed to identify the most important publications in the field (Batista-Canino et al., 2023). Thirdly, bibliographic coupling was applied to better capture emerging fields and research fronts (Boyack & Klavans, 2010), with a five-year timeframe was selected to ensure timeliness of the analysis (Clermont et al., 2021; Zupic & Čater, 2015) given the nature of rapid technology advancement. Finally, the current bibliometric analysis adopts the bioecological model to better examine and illustrate the dynamics of technology use in English language learning (Bronfenbrenner, 1979; Bronfenbrenner & Morris, 2006; Chiu, 2020; Navarro & Tudge, 2023). These approaches set the stage for the following research questions (RQ):

1. What is the current state of research in technology-enhanced English language learning in terms of performance analysis?
2. What are the most important publications based on the latest research in technology-enhanced English language learning?
3. What is the conceptual structure (research fronts) in technology-enhanced English language learning?
4. How can the field of technology-enhanced English language learning be contextualised in the bioecological theory?

Addressing RQ1 illustrates the overall scientific development of TELL in English language education. The core publications identified in RQ2 serve as a reference guide for scholars initiating research in the field. For RQ3, knowledge of the current and emerging research trends can further guide research topics and identify research gaps. Finally, framing the bibliometric findings within the bioecological model provides a holistic view of TELL in English language education.

Methods

Bibliometric analysis

Bibliometrics applies objective mathematical analysis on publications (Pritchard, 1969) and benefits from processing mass amounts of data (McBurney & Novak, 2002) and generating more accurate insights over conventional literature reviews (Cobo et al., 2011). Bibliometrics involves two types of analysis (Zupic & Čater, 2015). Performance analysis reveals the impact and production of authors, publications, journals, organisation, or countries. Scientific mapping visualises their relationships to construct the intellectual, conceptual, or social network structures of the field, and can be conducted using co-citation analysis, which clusters publication references based on them being cited together (Small, 1973), and bibliographic coupling, which clusters the publications by their citing the same references (Kessler, 1963). Both approaches assume connected publications share similar themes; whereas co-citation analysis is backward-looking and reveals knowledge base in the intellectual structure, bibliographic coupling is forward-looking and unveils the research fronts in the conceptual structure (Boyack & Klavans, 2010). Another approach to revealing the conceptual structure of the field is co-word analysis, which clusters the keywords that are linked according to the degree to which they co-occur in publications (Callon et al., 1991). Various tools have been used for bibliometric analysis, including VOSviewer (van Eck & Waltman, 2010) and R Bibliometrix (Aria & Cuccurullo, 2017).

Growing interest in bibliometrics has seen its application in language education. Scholars have reviewed second language acquisition (Zhang, 2019) and English as a medium of instruction (EMI; Wu & Tsai, 2022). Specific aspects such as listening and reading skills (Aryadoust, 2020), motivation in language learning (Wu, 2022), pre-school learners (Yilmaz et al., 2022, language learning in Southeast Asia (Ngoc & Barrot, 2023), and e-book usage in EFL (M. -R. A. Chen et al., 2021) have also been investigated.

The field of CALL has especially garnered ample attention. On a smaller scale, Goksu et al. (2022) examined 310 studies published between 2014 and 2019 to identify the most productive countries, organisations, and authors in the CALL journal. Thematically, they found English was the most common researched language, and that language skills were the most investigated dependent variable in investigating the impact of technology. This was supported by co-word analysis, which revealed CALL, MALL, EFL, blended learning, reading, writing, vocabulary, telecollaboration, and motivation as the most common keywords. Scholars have also utilised large datasets in their reviews. For instance, Lim and Aryadoust (2021) investigated the field of CALL from 1977 to 2020 by analysing 3,697 publications in 11 journals using co-citation analysis to identify the most impactful studies. The authors identified seven basic themes including computer-mediated communication and interaction, multimedia, telecollaboration or email exchanges, blogs, digital games, wikis, and podcasts.

In a comprehensive review combining bibliometrics with structural topic modelling on 1,295 publications from 1995 to 2019, X. Chen et al. (2021) found growing diversification and pedagogical application of technologies. Primarily, their analysis identified the increasing use of certain technologies (i.e. mobile, wikis, digital games, VR, etc.) and declining popularity of others such as digital books and multimedia content. Moreover, their results indicate diverse use of technologies in different contexts, such as mixed use of mobile technologies and glossaries for vocabulary learning and digital multimodal composing in project-based learning. In addition to revealing the lack of research on recent technological developments in artificial intelligence and learning analytics, the authors emphasised keeping abreast of latest technological trends and examining how they can be integrated into the language classrooms to foster better learning outcomes.

Five-step workflow of bibliometric analysis

The current bibliometric analysis follows Zupic and Čater (2015) five-step workflow, illustrated in Figure 1. First, according to the research questions, bibliographic coupling was chosen to identify the latest research trends and local citation was examined to identify the core publications in the field.

In Step two, search and retrieval of publications were conducted in accordance with PRISMA guidelines as employed in EFL technology review (Klímová & Seraj, 2023) and bibliometric analysis (Behl et al., 2022), with the search not limited by time scope and finished on May 28, 2023. The PRISMA flow

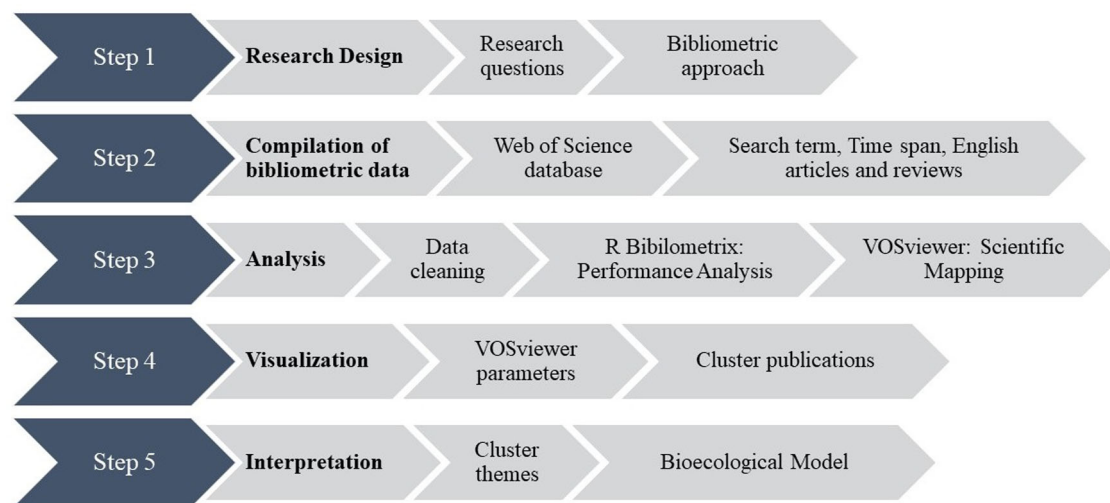


Figure 1. Five-step Workflow adapted from Zupic and Čater (2015).

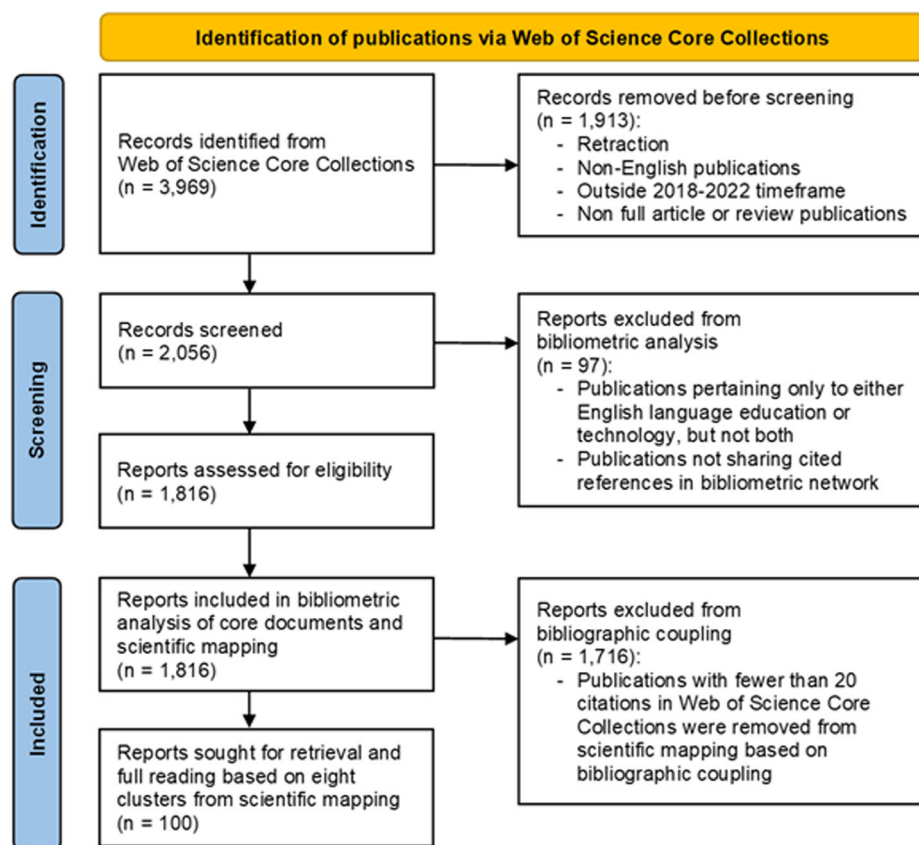


Figure 2. PRISMA flow diagram.

diagram is depicted in Figure 2. Identification of publications was conducted using the search terms in Table 1, which returned 3,969 results on Web of Science Core Collections. The dataset was then filtered to include only English language articles and reviews (including early access) published between 2018 and 2022. The full five-year timeframe was chosen in order to best identify emerging fields and smaller subjects using bibliographic coupling (Clermont et al., 2021; Zupic & Čater, 2015). This resulted in 2,056 documents exported and downloaded for screening.

Step three includes data cleaning, performance analysis, and scientific mapping. Data cleaning involved filling in missing publication years through DOI search. For performance analysis, scientific

Table 1. Search query.

English language education terms	TS=(“english as an international language” OR “EIL” OR “english as medium of instruction” OR “EMI” OR “english language teach*” OR “ELT” OR “english as a second language” OR “ESL” OR “english as a foreign language” OR “EFL” OR “teaching English as a foreign language” OR “TEFL” OR “teachers of english to speaker* of other language*” OR “TESOL” OR “learn English” OR “teach English” OR “English educat*” OR “English learn*” OR “English teach*” OR “English instruct”)
Technology-related terms	AND TS=(“flipped learn*” OR “flipped class*” OR “blended learn*” OR “blended class*” OR “hybrid learn” OR “hybrid class*” OR “synchronous class*” OR “asynchronous class*” OR “synchronous teach*” OR “synchronous learn*” OR “asynchronous teach*” OR “asynchronous learn*” OR “technology-enhanced” OR “technology-mediated” OR “digital learn*” OR “digital technolog*” OR “digital education” OR “e-learning” OR “electronic learn*” OR “elearning” OR “information and communication technolog*” OR “ICT” OR “distance learn*” OR “distance educat*” OR “distributed learn*” OR “remote learn*” OR “remote teach*” OR “web 2.0” OR “web-enhanced learn*” OR “web-enhanced instruct*” OR “web-based instruct*” OR “web-based train*” OR “web* teach*” OR “web* learn*” OR “internet* learn*” OR “internet* teach*” OR “online learn*” OR “online educat*” OR “online teach*” OR “online pedagog*” OR “online instruct*” OR “massive open online course*” OR “MOOC” OR “virtual class*” OR “virtual learn*” OR “virtual instruct*” OR “virtual teach*” OR “virtual learning environment*” OR “learning management system*” OR “LMS” OR “course management system*” OR “CMS” OR “computer-based learn*” OR “computer-based instruct*” OR “computer-supported learn*” OR “computer-mediated learn*” OR “computer-assisted language learn*” OR “CALL” OR “mobile learn*” OR “m-learning” OR “mLearning” OR “mobile assisted language learn*” OR “MALL” OR “mobile comput*” OR “computer mediated communicat*” OR “CMC” OR “digital game-based”)

development and core publications were revealed using the R-Bibliometric/Biblioshiny package (Aria & Cuccurullo, 2017). Core publications were identified using local citation rather than global citation (Batista-Canino et al., 2023). This is because publications with high global but low local citations would mean that they are not pertinent to the field under investigation, while high local citation counts, regardless of global citation counts, reveal field-specific publications. As such, local citation count was used to identify the core publications.

Scientific mapping was generated through the VOSviewer software (van Eck & Waltman, 2010). VOSviewer provides distance-based visualisation involving three steps: normalisation, mapping, and clustering (van Eck & Waltman, 2014). VOSviewer first applies association strength normalisation to account for the disparity between highly cited publications than less cited ones in the dataset. Then mapping is created by positioning of the nodes, or publications, in two-dimensional space. Close distances signify high relatedness, and vice versa. Finally, VOSviewer uses a smart local moving algorithm to cluster closely related nodes, with nodes assigned to distinct coloured clusters.

For the visualisation in step four, minimum citations of 10, 20, and 30 were used in VOSviewer to generate mappings to further identify publications not related to both English language education and technology. Through reading the abstracts, or if necessary, the entire publication, this further removed 97 publications, with a final dataset comprising 1,816 publications used for visualisation. The minimum citation of 20 was set, as it best generated clusters with similar research themes. The default VOSviewer cluster colours were used, with red being the largest cluster, followed by green, blue, yellow, purple, teal, orange, and brown. Interpretations during the final step involved close reading of the publications in the clusters to identify their themes.

Results and discussion

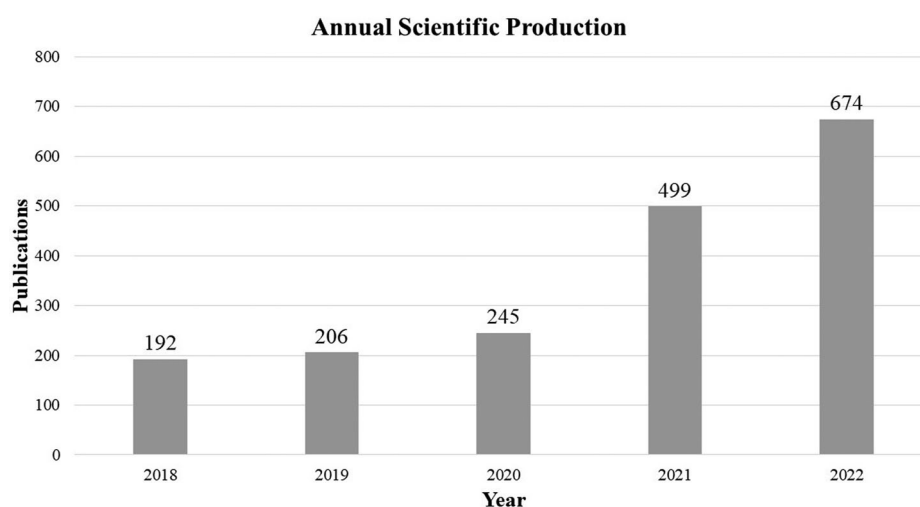
A descriptive summary of the publication dataset including the main information about the data, authors and author collaborations, and document contents and types, is provided in Table 2. The final dataset comprised 1,816 articles and reviews published between 2018 and 2022, which included early access publications available in 2022. Figure 3, which depicts the annual scientific production, shows continual growth with an annual growth rate of 31.66%.

Core publications

In Table 3, the top ten core publications, listed based on the local citation counts, are the most highly cited studies within the 1,816 publications. Wherein there is a tie between two publications, normalised

Table 2. Publications dataset summary.

Description	Results
MAIN INFORMATION ABOUT DATA	
Timespan	2018:2022
Sources (Journals, Books, etc)	405
Documents	1,816
Annual Growth Rate %	31.66
Document Average Age	2.4
Average citations per doc	5.167
References	58,308
AUTHORS	
Authors	3,368
Authors of single-authored docs	463
AUTHORS COLLABORATION	
Single-authored docs	538
Co-Authors per Doc	2.33
International co-authorships %	16.74
DOCUMENT CONTENTS	
Keywords Plus (assigned by WoS)	1,317
Author's Keywords	4,147
DOCUMENT TYPES	
Articles	1,745
Reviews	71

**Figure 3.** Annual scientific production.

local citation is used, followed by the global citation. The core publications show that recent research in the field of TELL in English language education has demonstrated strong focus on flipped classroom models and technology-enhanced learning. For instance, Turan and Akdag-Cimen (2020)'s systematic review of 43 articles found that the flipped classroom method in ELT gained significant popularity after 2014, with a rapid increase in studies focusing on this approach. This method, often incorporating mixed and quantitative research methodologies, has been particularly effective in improving speaking and writing abilities in EFL contexts. Similarly, Lin and Hwang (2018) meta-analysis of 63 experimental articles confirmed that flipped classrooms can enhance students' academic performance in EFL settings. Overall, students demonstrated higher scores and increased engagement (Lee & Wallace, 2018), supporting the effectiveness of this model in EFL learning. Beyond flipped classrooms, Lin and Lin (2019) highlighted the positive impact of MALL on vocabulary retention, with messaging services showing more efficacy than mobile applications. Nevertheless, successful vocabulary acquisition requires a diverse range of high-quality digital learning experiences is more beneficial than merely the quantity of such activities (Lee, 2019). Teachers must also overcome the challenges and necessary adaptations in shifting towards more technologically integrated teaching methods in various EFL contexts (Gao & Zhang, 2020; Lee & Wallace, 2018; Lin & Hwang, 2018; Lin & Lin, 2019; Turan & Akdag-Cimen, 2020).

Table 3. Core publications.

Rank	Author	Title	Year	Local citations	Global citations	Normalised local citations
1	Z. Turan & B. Akdag-Cimen	Flipped classroom in English language teaching: A systematic review	2020	41	97	21.25
2	G. Lee & A. Wallace	Flipped Learning in the English as a Foreign Language Classroom: Outcomes and Perceptions	2018	37	80	18.60
3	C. -J. Lin & G. -J. Hwang	A learning analytics approach to investigating factors affecting EFL students' oral performance in a flipped classroom.	2018	33	66	16.59
4	A. E. P. Atmojo & A. Nugroho	EFL classes must go online! Teaching activities and challenges during COVID-19 pandemic in Indonesia	2020	26	61	13.48
5	J. -J. Lin & H. Lin	Mobile-assisted ESL/EFL vocabulary learning: a systematic review and meta-analysis	2019	26	69	12.34
6	M. Amiryousefi	The incorporation of flipped learning into conventional classes to enhance EFL learners' L2 speaking, L2 listening, and engagement	2019	24	37	11.39
7	D. Zou & H. Xie	Flipping an English writing class with technology-enhanced just-in-time teaching and peer instruction	2019	20	46	9.49
8	B. Mei, G. T. L. Brown & T. Teo	Toward an understanding of preservice English as a foreign language teachers' acceptance of computer-assisted language learning 2.0 in the People's Republic of China.	2018	17	67	8.54
9	H. Haghighi, M. Jafarigohar, H. Khoshima, & F. Vahdany	Impact of flipped classroom on EFL learners' appropriate use of refusal: Achievement, participation, perception	2019	15	35	7.12
10	J. S. Lee	Informal digital learning of English and second language vocabulary outcomes: Can quantity conquer quality?	2019	15	26	7.12

Scientific mapping

The conceptual structure of technology in English language education using bibliographic coupling of the 1,816 publications with more than 20 citations generated eight clusters comprising 100 publications, as depicted in [Figure 4](#). Each cluster represents a group of nodes (publications) with highly connected themes. The size of the node is indicative of the publication's total link strength, or its degree of connectivity and influence within the network. Positionally, clusters or nodes closer to the centre represent core and well-established themes, while those further from the centre are niche or emerging themes. The following findings for each cluster theme are arranged according to their cluster sizes (number of nodes).

Red cluster (1): COVID-19

As the largest cluster, the publications revealed how the pandemic forced English teachers to implement emergency distance education (Huang et al., 2021; Karataş & Tuncer, 2020; Sepulveda-Escobar & Morrison, 2020) without sufficient training or experience (Atmojo & Nugroho, 2020; Marshall et al., 2020). The virtual nature further aggravated their teaching, making it difficult to enforce student compliance (Oraif & Elyas, 2021), monitor student performance (Gao & Zhang, 2020), and hold students accountable or motivated (Marshall et al., 2020). Students, on the other hand, experienced high boredom and distracted themselves due to the non-interactive nature, high cognitive demand, and superficiality of online classes (Pawlak et al., 2021).

Nevertheless, the experience allowed English teachers to learn different technology platforms, design strategies, and ways to interact with students (Moorhouse et al., 2021; Sepulveda-Escobar & Morrison, 2020). It identified and highlighted the importance technology competencies (e.g., flexibility in adopting different tools and familiarity with tools and features), online environment management competencies (e.g., designing lessons that maintain student-teacher interaction that accounts for environment limitation) and online teacher interactional competencies (e.g., using multiple modes of communication and questioning techniques) in online education (Atmojo & Nugroho, 2020; Moorhouse et al., 2021;). It also drew attention to the digital divide (Atmojo & Nugroho, 2020) and educational inequalities (Gao &

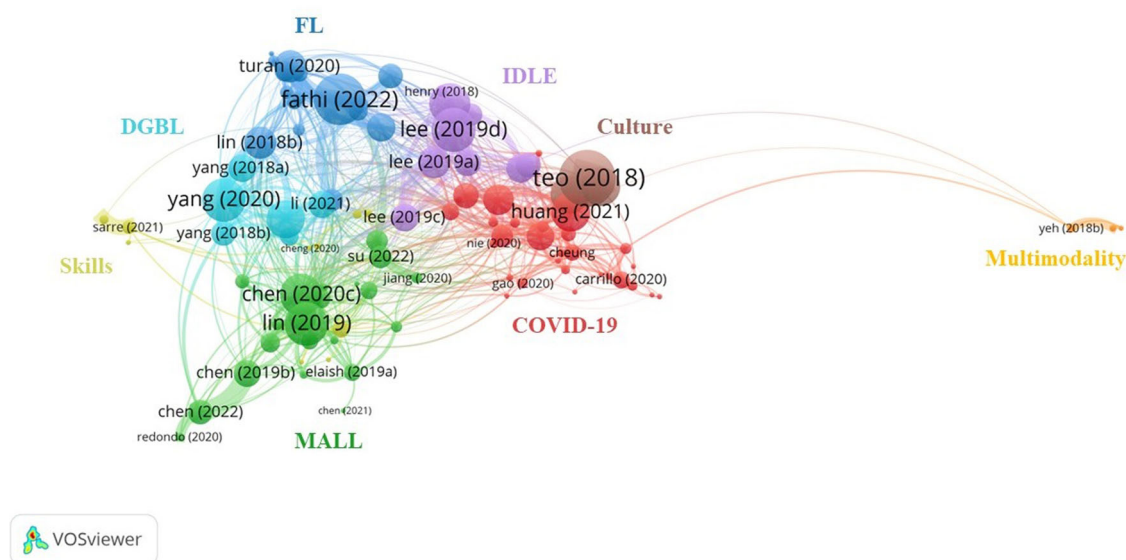


Figure 4. Scientific mapping of TELL research themes.

Zhang, 2020; Karataş & Tuncer, 2020). In short, successful online course implementation depended on the teachers' role and support (Karataş & Tuncer, 2020), clear understandings of students' learning needs (Gao & Zhang, 2020), and their reflecting on teaching practices (Mumford & Dikilitaş, 2020).

Additionally, utilizing Davis's (1989) technology acceptance model (TAM) in the pandemic context, the cluster provided further support for the antecedents of perceived ease of use (PEOU; e.g., whether technology requires little effort) and perceived usefulness (PU; e.g., whether technology facilitated teaching or learning) to behavioural intentions (BI) of students (Abraham et al., 2019; Fathali & Okada, 2018; Li et al., 2019) and teachers (Bai et al., 2019; Huang et al., 2021; Rafiee & Abbasian-Naghneh, 2019) to adopt technology. Student and teacher PEOUs were also positively impacted by computer self-efficacy (Bai et al., 2019; Li et al., 2019; Rafiee & Abbasian-Naghneh, 2019), and negatively affected by anxiety (Bai et al., 2019; Li et al., 2019). For teachers, facilitating conditions (FC) of administrative and technology support and access (Huang et al., 2021) and technology interest and help-seeking attitude in learning ICT (Bai et al., 2019) contributed to technology use, while fulfillment of the basic psychological needs of competence, autonomy, and relatedness contributed to students' PU and PEOU (Fathali & Okada, 2018). Lastly, students' acceptance of technology was also investigated through push-pull-mooring-habit framework (Chen & Keng, 2019), theory of planned behaviour (Nie et al., 2020), and course satisfaction (Bailey et al., 2021), all of which exerted positive impact on acceptance.

Green cluster (2): MALL

One of the key findings in the green cluster was overwhelming research interest and the popularity and positive impact of MALL for vocabulary acquisition (T. Chen et al., 2021; Z. Chen et al., 2020; Elaish et al., 2019; Lin & Lin, 2019; Poláková & Klímová, 2019; Zhang & Pérez-Paredes, 2019). Meta-analyses have also found positive and medium-to-large effect sizes for speaking, listening, writing, and vocabulary learning (Lin & Lin, 2019; Z. Chen et al., 2020).

Researcher-designed tools based on various theoretical frameworks were also proposed. For instance, Hao et al. (2019) vocabulary learning app incorporated the cognitive apprenticeship model and featured listening, speaking, reading, and writing lessons to enhance the confidence and attitude toward English learning of struggling students. Personalised recommendations based on technique feature analysis utility and task diversity also maximised vocabulary learning outcome (Zou & Haoran, 2018). Introduction of game-related functions based on principles of persuasive technology improved performance even for students with low motivation (Elaish et al., 2019), while gamified assessment with ranking and competition mechanisms improved vocabulary acquisition, retention, and engagement compared to those without (C. -M. Chen et al., 2019).

Another major finding was the shift towards situated and collaborative learning (Z. Chen et al., 2020; Chung et al., 2019; Kaceti & Klímová, 2019; Klímová, 2017; Shadiev et al., 2019; Su & Zou, 2020), both of which demonstrated high effect sizes (Z. Chen et al., 2020). On the one hand, collaborative learning through social media and web 2.0 tools such as instant messaging services promoted student engagement, particularly benefitting low achievers and shy students, who can choose when and how to participate (Klímová, 2017; Su & Zou, 2020). Reduced anxiety led to increased motivation, greater collaboration, and effective learning. Even students unfamiliar with each other benefitted from explicit socialising activities using *Wechat*, which facilitated their level of social presence and acquisition of complex cognitive skills in essay writing (Jiang & Zhang, 2020).

On the other hand, situated learning allows students to apply their language knowledge in authentic real-world environments. For instance, in addition to enhancing student learning, use of augmented reality enhanced self-efficacy, learning value, and proactive learning (M. -P. Chen et al., 2019) and students' socio-affective relationships (Redondo et al., 2020). However, for novice or inexperienced learners, situated learning could be too complex, resulting in high cognitive load (Chung et al., 2019; Shadiev et al., 2019). In this regard, offering teachers' guidance and assistance (Chung et al., 2019; Su & Zou, 2020), reducing extraneous information during learning (M. -P. Chen et al., 2019, 2020), providing multimodal learning activities (Zhonggen et al., 2018), and implementing collaborative learning (Jiang & Zhang, 2020; Su & Zou, 2020) can reduce cognitive load and lead to better learning outcomes.

Blue cluster (3): FL

Publications in the blue cluster provided support for positive impact of flipped learning (FL) on learning outcome in the four skills (Lee & Wallace, 2018), including writing (Fathi & Rahimi, 2020; Lin et al., 2018; Su Ping et al., 2019; Turan & Akdag-Cimen, 2020; Wu et al., 2019; Zou & Xie, 2018), speaking (Abdullah et al., 2019; Amiryousefi, 2017; Chen & Hwang, 2019; Haghighi et al., 2019; Lin & Hwang, 2018), listening (Amiryousefi, 2017; Chen & Hwang, 2019; Turan & Akdag-Cimen, 2020), and grammar (Liu et al., 2018). Students experienced higher self-efficacy by having more time for pre-class preparation and more in-class interaction and feedback (Namaziandost & Çakmak, 2020; Su Ping et al., 2019). Affectively, FL classes were not only more enjoyable and engaging (Haghighi et al., 2019; Lee & Wallace, 2018; Su Ping et al., 2019) but also circumvented students' demotivation factors by maintaining positive self-interest, classroom atmosphere, learning content, and language output (Wu et al., 2019). In short, FL provided both affective and cognitive advantages when compared with conventional teaching methods (Fathi & Rahimi, 2020; Haghighi et al., 2019; Lee & Wallace, 2018; Lin et al., 2018; Lin & Hwang, 2018; Namaziandost & Çakmak, 2020) and has become a new trend in English courses (Hockly & Dudeney, 2018).

In terms of in- and out-of-class activities, the FL model can also be enhanced with other technologies and pedagogical designs. Primarily, students perceived mobile devices as useful and easy to use given their high familiarity (Andujar et al., 2020; Lin & Hwang, 2018; Liu et al., 2018). Used as student response systems (SRS) to input their responses in class, mobile learning platforms such as Peardeck enhanced student motivation and engagement (Liu et al., 2018) while game-based SRS such as Kahoot! further benefited their learning skills and confidence (Zou, 2020). Pedagogically, in-class activities supported with concept maps, or mind maps of key ideas and their relationships, were more advantageous than conventional worksheets in improving students' proficiency level, critical thinking awareness, and reducing speaking anxiety (Chen & Hwang, 2019).

Multiple technologies also supported out-of-class activities. Rather than receiving lectures and playing games during class, watching instructional videos and playing the game before class better reduced writing errors, promoted positive student reflections in perceived usefulness, motivation, and satisfaction towards the game, and performance (Lin et al., 2018). Out-of-class activities integrating collaborative writing and note-taking through cloud-based tools such as Google Docs and Padlet also provided scaffolded learning and sharing opportunities amongst peers, thereby promoting in-class peer instruction and just-in-time teaching (Zou & Xie, 2018). On the other hand, student cooperation improved students' out-of-class engagement with materials and activities preparation in listening and speaking classes (Amiryousefi, 2017). Using Facebook as an out-of-class FL platform for watching instructional video

learning content and sharing what was learned similarly enhanced students' oral performance (Lin & Hwang, 2018).

Despite positive findings, the studies also raised some FL issues. Turan and Akdag-Cimen's (2019) review of 43 FL studies revealed that at most, FL benefitted student engagement, warranting future meta-analysis to better ascertain its impact on learning outcome. However, successful in-class engagement requires students' pre-class learning (Zou, 2020), and the extra workload may result in lower satisfaction and engagement (Amiryousefi, 2017; Turan & Akdag-Cimen, 2020; Zou, 2020). Inability to complete tasks independently also resulted in anxiety (Lin et al., 2018) while out-of-class preparation was perceived as boring and time-consuming (Su Ping et al., 2019).

Yellow cluster (4): skills

The yellow cluster reveals how technology benefits students' particular language skills. In terms of writing, providing students with overall performance comments on organisation, grammar, and vocabulary in addition to corrective feedback (CF) targeting grammar errors reduced students' subsequent error rates than those without CF (Sarré et al., 2019). Amongst six CF types, unfocused (all errors identified) and indirect (no correction provided but includes metalinguistic comment provided to identify the nature of errors) CF accompanied by extra computer-mediated grammar drills showed the highest decrease in error rate. Teacher's asynchronous electronic feedback could be further complemented by synchronous text-based feedback chat sessions, providing opportunities to reinforce prior feedback and address students' needs and higher-order concerns rather than language structures (Ene & Upton, 2018). Students themselves could also employ machine translations such as Google Translate, which particularly improved vocabulary accuracy and sentence structures of lower-lower students (Lee, 2019).

Another sub theme of this cluster is English listening and speaking skills through different aspects of technology-enhanced education. In terms of activity design, a nine-stage focus-on-form (FonF) model was found to enhance student listening proficiency and benefited students' speaking confidence and anxiety (Bahari, 2019). The model allowed learners to choose their own learning materials, thereby reinforcing their self-confidence, self-efficacy, and self-determination. Within the FonF model, three stages focused on form (e.g., mimicking body movements and facial expressions), meaning (e.g., repeating and rehearsing), and communication (e.g., producing language output), which minimised cognitive load by task repetition in the early stages and facilitating form and meaning connection prior to communicative output. In terms of course design, Wang et al. (2019) provided a blended learning model that supports face-to-face interactive learning via flipped classroom, online learning community through small private online courses (SPOC), individual learning via mobile devices, and out-of-class language use via project-based learning. The results showed that students not only perceived it as effective and motivated them to learn and use English, but that the use of SPOC provided a natural context to develop learner autonomy and improved participation and satisfaction with the course. Finally, in terms of tool use, students found the use of intelligent personal assistants, Alexa, both enjoyable and useful, though positive learning outcomes were found only for listening and not speaking (Dizon, 2020). On the other hand, Yilmaz et al.'s (2022) bibliometric review identified the most preferred technologies by teachers to include educational robots, digital games, mobile technologies, music/voice, and CD-ROMs. Design of educational robots, however, requires comprehensive design and evaluative framework and cross-disciplinary expertise (Cheng et al., 2020). For positive interaction with learners, educational robots must feature human-like linguistic feedback (verbal and non-verbal), provide adaptable content for different learners, engage learners' body movements, and utilise an authoring interface that is user friendly and easy to learn. In sum, future research should focus more on pedagogical aspects and whether technology use contributed to EFL education (Yilmaz et al., 2022).

Purple cluster (5): IDLE

The purple cluster focused on informal digital learning of English (IDLE), which appeared most prominently (Lee, 2019b, 2019c; Lee & Drajiati, 2019; Lee & Lee, 2020). Related terms also included extramural digital content (Lee, 2019a) and online informal learning of English (Lamb & Arisandy, 2019). Referring to students' autonomous English learning outside the classroom using unstructured online digital environments without formal teacher instructions or courses, IDLE exerted positive impacts on both emotions

(Lamb & Arisandy, 2019; Lee, 2019c; Lee & Lee, 2020;) and learning outcomes (Lee, 2019b, 2019c). Students who practised IDLE activities (e.g., watching English YouTube videos or interacting with English speakers on social media) more frequently experienced greater enjoyment (Lee, 2019c; Lee & Lee, 2020) and confidence (Lamb & Arisandy, 2019; Lee, 2019c) in English. English confidence and enjoyment were also significantly correlated with the quantity of IDLE activities (Lee, 2019c). While IDLE quantity did not impact learning outcome (Lee, 2019b), the diversity of IDLE activities was found to both positively impact confidence in addition to speaking (Lee, 2019c) and vocabulary acquisition (Lee, 2019b, 2019c). For instance, utilising English digital resources to learn or memorise English words, watching entertainment programs, and communicating via social media altogether contributed to productive language outcomes. Compared to entertainment or self-instructive purposes however, socially-oriented IDLE activities were the least reported activity (Lamb & Arisandy, 2019).

In this regard, students' social engagement in IDLE contexts was found to be a consequence of socio-political (e.g., K-12 instructions), contextual (e.g., familiarity with others and communities), and individual variables (e.g., English self-confidence and anxiety), which either hindered or promoted their willingness to communicate (WTC; Lee, 2019a). In particular, socio-political factors such as K-12 teacher-centred instructions and test-oriented curriculum have resulted in students being accustomed to avoiding or minimising English communication both inside and outside the classroom. Conversely, having close foreign friends or strong familiarity with the virtual community facilitated their WTC in the IDLE context (Lee, 2019a; Lee & Drajeti, 2019). In fact, productive-oriented IDLE activity was found to be a significant predictor of WTC both inside and outside English classes. Thus, given the positive learning impact of communicative IDLE activities, EFL teachers' instruction should leverage classroom activities to enable students to experience closer fit with their leisure-time discourse and motivate their IDLE engagement (Henry et al., 2018).

Teal cluster (6): DGBL

The teal cluster provided support for the positive impact of digital game-based learning (DGBL) in English classes. For instance, Xu et al. (2020) scoping review of 59 publications on DGBL technologies in English learning found nearly 80% ($n=47$) reported positive impact on language acquisition, with games providing a scaffolding experience for students. Learning was promoted through repeated exposure to gaming environment contents (Chen & Hsu, 2020) or repeated practice given the desire for rewards (Yang et al., 2018a). While this benefited students with lower proficiency and decreased their anxiety, if they could not understand in-game English instructions or tasks due to language difficulties, they experienced higher anxiety and subsequently lower performance improvement compared to students with higher English proficiency (Yang & Quadir, 2018a). To promote optimum learning outcomes, games should provide balanced skill and challenge with clear goal and playability to keep students motivated through flow experience (Li et al., 2021) while accounting for user performance (Yang et al., 2020), gender (Yang & Quadir, 2018b), and prior gaming experiences (Yang & Quadir, 2018a).

Orange cluster (7): Multimodality

The orange cluster is entitled "Multimodality", referring to the channels of language transaction, notably in terms of receptive and productive languages and the four skills of reading and listening (receptive) and speaking and writing (productive). Primarily, Grapin (2018) draws attention to the contrast between weak multimodalities limited to linguistic modalities in language development against the strong multimodalities in other content disciplines (e.g., mathematics, sciences, etc). In turn, the author argues for the shift toward strong multimodalities, which leverages the unique affordance and overcomes the limitations of different modalities (e.g., image and text), creating greater meaning-making for language learners. In this regard, digital multimodal composing (DMC) leveraging advances in technology has emerged to engage students with different multimedia modalities to nurture their English proficiencies across the four skills and meaning-making (Jiang & Ren, 2020; Yeh, 2018). On the one hand, students positively perceived DMC to benefit not only their vocabulary, speaking, and translation and writing, but also allowed them to learn about their own culture and gain multimedia skills (Yeh, 2018). Accessing online video resources further provided other non-linguistic modalities such as body language and gestures to help students better act out their script and communicate their ideas. Nevertheless, contrasting

ideologies between the student and teachers may arise in DMC pedagogy over the nature of language, the teachers' role, and evidence of learning (Jiang & Ren, 2020). Specifically, teachers' insistence on adhering to linguistic systems and curricular-based evaluations hindered students' investment in language learning during DMC. Thus, English language teachers must move beyond conventional weak multimodalities toward stronger multimodalities to promote language development (Grapin, 2018).

Brown cluster (8): culture

The brown cluster is entitled "Culture" and features three studies that highlight the unique facilitating conditions of the Chinese cultural, national, and organisational contexts to technology adoption. Primarily, technology use was promoted by the collectivist Confucian-heritage, which strengthened the influences of compulsory mandates and group interests (Teo et al., 2018). As such, school policy and organisational culture significantly impacted their behavioural intention to use technology (Huang & Teo, 2020). Rather than expressing concerns, complaints, or resentment, Chinese teachers further took the initiative to use technology and were able to transform their perception from technology as time-consuming to one of time-saving pedagogy (Huang et al., 2019).

Conclusion

This study investigated and identified the current state-of-the-art research in English TELL through bibliometric approaches. Performance analysis (RQ1) reveals continuing exponential growth of technology use in English learning. The core publications (RQ2) centred on FL and provides a reference for key publications in this field for scholars. Scientific mapping of the conceptual structure using bibliographic coupling revealed eight research fronts (RQ3). MALL has become the prominent technology in TELL, with FL and DGBL as dominant pedagogical approaches. In terms of English skills, TELL mainly supported teaching of writing skills. This study further revealed technology multimodality and informal learning of English as emerging themes for future research. The COVID pandemic further underscored the importance of supporting teachers in technology use. In the following, these findings are framed within the bioecological model, illustrated in Figure 5, to provide a comprehensive overview and recommendations for the field of TELL in English language education to answer RQ4.

Student as core of bioecological model

As the focal point of the bioecological model (Bronfenbrenner, 1979; Bronfenbrenner & Morris, 2006), students' English outcome in vocabulary (Lin & Lin, 2019), writing (Ene & Upton, 2018), and listening and speaking (Bahari, 2019; Wang et al., 2019;) are impacted by each of the systems within. Successful implementation of TELL must further consider students' trait-specific factors such as gender (Yang & Quadir, 2018b), English proficiency (Yang & Quadir, 2018a), motivation (Haghighi et al. 2019; Lee & Wallace, 2018; Wu et al., 2019), and self-regulation (Zou, 2020). While students are digital natives familiar with and receptive to technology (Andujar et al., 2020; Lin & Hwang, 2018; Liu et al., 2018), acceptance in the learning context is still impacted by students' ICT/computer self-efficacy (Zou, 2020), attitude (Lin & Hwang, 2018; Liu et al., 2018), and PE/PEOU toward the system (Abraham et al., 2019; Fathali & Okada, 2018; Li et al., 2019). Educators must therefore not assume TELL will always be easily adopted by students; guidance and training remain crucial to overcoming students' inexperience with innovative pedagogical approaches to promote positive TELL outcomes.

Microsystem, techno subsystem, and mesosystem: Technology-enhanced language learning

Students' learning experiences are directly impacted by the microsystem and mesosystem, both of which are enhanced by the techno-subsystem. The current study shows that TELL research has primarily focused on MALL (red cluster) as the dominant technology in the techno-subsystem. FL (blue cluster) and DGBL (teal cluster) have been two prominent pedagogical approaches leveraging technology capabilities that enhance student-student, student-teacher, and student-content interactions across environmental contexts in the microsystem.

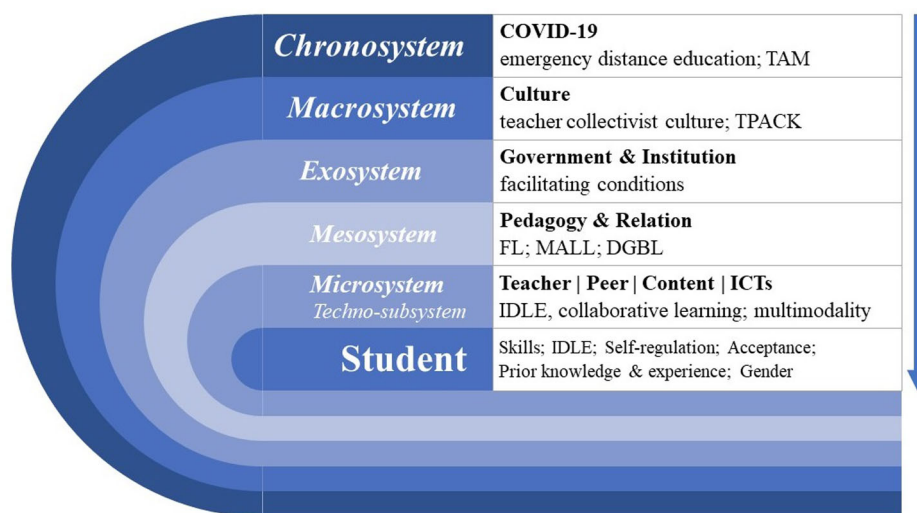


Figure 5. Bioecological model for TELL.

From the perspectives of the bioecological model, TELL research has blurred the inter- and intra-system boundaries, or the virtual and physical spaces. In terms of the techno-subsystem, FL leverages multiple web 2.0 technologies such as instant messaging services (Jiang & Zhang, 2020), social networking sites (Klímová, 2017; Su & Zou, 2020), and DGBL such as game-based student response systems (Amiryousefi, 2017; Liu et al. 2018; Zou, 2020) to improve teacher feedback and peer interaction. Out-of-class student-content interaction can support collaboration learning (Lin & Hwang, 2018; Zou & Xie, 2018), while in-class use of AR through mobile devices promotes students' socio-affective relationships (Redondo et al., 2020). Multimodality (orange cluster), involving use of digital multimedia composing through multiple technological tools and resources facilitates cultivation of the four skills and meaning-making (Grapin, 2018) has become a new area of TELL research. Beyond the confines of formal education and instructions, the emergence of IDLE (purple cluster) further reveals the importance of students' autonomous learning of English. TELL research shifts the emphasis on technologies (techno-subsystem) and unidirectional relationships (microsystem) toward multidirectional interactions between students, peers, teachers, and contents across both formal and informal and virtual and physical learning contexts. Thus, future research should examine the teachers' role and competency in integrating content delivery and technology use in TELL to enhance students' collaborative and autonomous learning.

Chronosystem, macrosystem, and exosystem: Trickle-down repercussion of the COVID-19 pandemic

Situated in the outermost system, the COVID pandemic (red cluster) shows how events in the chronosystem can exert far-reaching repercussions through all systems, especially for teachers and students. In terms of the impact on the macrosystem, the current study reveals that teachers in a collectivist culture may better alleviate technological challenges. In the exosystem, governments must first confront issues of digital divide (Atmojo & Nugroho, 2020) and educational inequalities (Gao & Zhang, 2020; Karataş & Tuncer, 2020) for students. Promoting curriculum policy grounded on constructivist beliefs can also better incentivize teachers' technology use (Teo et al., 2018). More directly connected to teachers and students in the exosystem is the institution, which plays a key role in providing the facilitating conditions to alleviate teachers' unwillingness to use technologies. Stemming from technical difficulties and access (Alfalah, 2018; Gönen, 2019) and lack of technology knowledge due to time constraints and anxiety (Taghizadeh & Hasani Yourdshahi, 2020), administrative policy and technical support must be provided to assist teachers (Huang et al., 2021; Huang & Teo, 2020; Mei et al., 2018). Providing hands-on training and practicums can help teachers realise that the time and effort required to learn technology could actually save them more time in the long run. Incorporating design-thinking in their training could further benefit teachers' technology, pedagogy, and content knowledge (TPACK), motivating them to explore different solutions and enhance their creativity in using technology (Mei et al., 2018; Tseng et al., 2019). Institutions must therefore ensure

the connection between school culture, professional development, and teacher knowledge to result in a self-perpetuating cycle of technology integration (Cheung, 2023).

Takeaways and limitations

This study has comprehensively explored the current landscape of English TELL through bibliometric analysis, highlighting the exponential growth and evolving dynamics within this field. The findings revealed the blurring of the physical and virtual pedagogical approaches wherein technology comprises all aspects such as mobile-assisted language learning (green cluster), flipped learning (blue cluster), and digital game-based learning (teal cluster). These approaches have significantly shaped the way English language skills (yellow cluster), particularly writing, are taught and learned. The investigation also revealed critical emerging themes such as technology multimodality (orange cluster) and informal digital English learning (purple cluster), suggesting new directions for future research. Framed within the bio-ecological model is a holistic view of the interaction and key considerations between and within each system, providing educational stakeholders a reference as to align technology adoption that accentuates student-centred approaches.

However, this study is not without limitations. Firstly, it focuses primarily on journal articles and reviews, thus excluding other types of academic publications such as conference papers, books, and reports, which might offer additional insights into the field. Secondly, the publications were drawn solely from the Web of Science database, limiting the breadth of the research, as incorporating other databases like Scopus or the grey literature could have provided a more comprehensive overview. Thirdly, by restricting the analysis to publications in English, the study potentially overlooks significant contributions in other languages, which could offer different perspectives or insights. Lastly, the exclusive use of bibliographic coupling as the bibliometric approach, while effective, limits the methodological scope. Employing other bibliometric methods such as co-word analysis, could provide a more nuanced understanding of the field's development and current state. These limitations suggest avenues for further research to build upon and expand the findings of this study.

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Data availability statement

The data that support the findings of this study are available from *Clarivate's Web of Science Core Collection*. Restrictions apply to the availability of these data, which were used under university database access licence for this study. Data is available from the author(s) with the permission of *Clarivate*.

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